CEC 315: Signals and Systems

The mathematical foundation of modern engineering systems

COURSE ESSENTIALS

What You Need to Know

Prerequisites

EGR 115

MA 345

Mathematical maturity required

Abstract thinking essential

Course Structure

Mini-lectures + active problem solving

Primary text: Oppenheim, Willsky, Nawab

Homework drives learning

Office hours: TBD

INSTRUCTOR

Dr. Rogelio Gracia Otalvaro

Rogelio Gracia Otalvaro received his Ph.D. in Electrical Engineering and Computer Science at Embry-Riddle Aeronautical University, where he served as a Graduate Research and Teaching Assistant. He holds master's degrees in Industrial Technologies Engineering and Mechanical Engineering, and a bachelor's degree in Industrial Technologies Engineering from ICAI Universidad Pontificia de Comillas, Madrid. His research integrates dynamical systems theory with ecological network analysis to enhance the resilience of critical infrastructure. His publications and conference presentations at venues including INCOSE and ASME span power systems analysis, machine learning in aerospace, and bio-inspired design.

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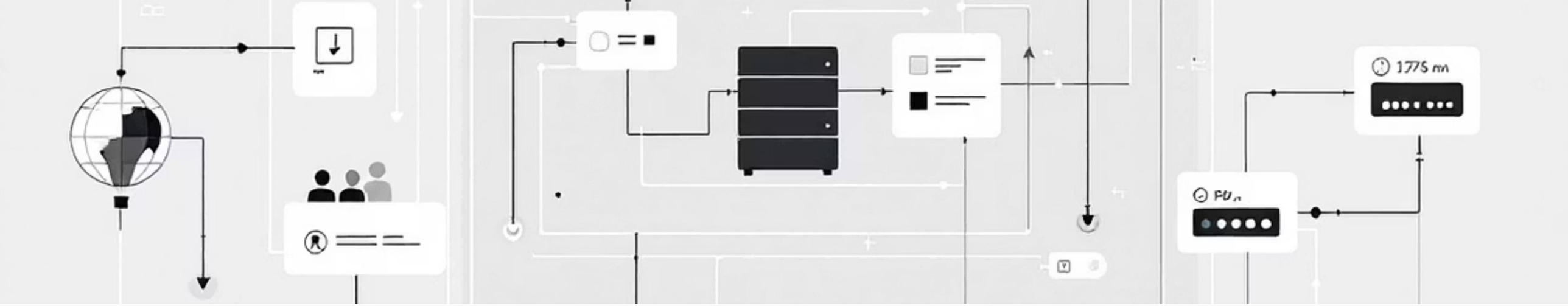
Virtual office hours Tuesdays and Thursdays from 10:00–11:00 a.m. – Please stop by!

BID4R

Biologically Inspired Design-for-Resilience Lab



Why This Course Matters

Universal Language

Every engineered system speaks signals: input → process → output

Predictive Power

Model behavior mathematically before building hardware

Engineering Limits

Understand bandwidth, stability, distortion fundamentally

Where Signals & Systems Live

Communications

Filters, modulation, bandwidth allocation. How information travels wirelessly

Control Systems

Stability analysis, impulse response. Keeping aircraft, robots, and factories stable

Signal Processing

Audio compression, image filtering, radar detection. Extracting meaning from noise

Embedded DSP

Real-time processing on hardware. Smartphones, sensors, navigation systems

The Intellectual Journey

1. **Signal Representation**
   Time domain vs frequency domain perspectives
2. **System Behavior**
   How systems transform inputs into outputs
3. **Convolution Power**
   The fundamental operation connecting signals and systems
4. **Frequency Tools**
   Simplifying complex behavior through spectral analysis
5. **Discrete Meets Continuous**
   Bridging analog and digital worlds seamlessly

FRAMEWORK THINKING

This Is a Foundational Course

> You're learning a framework that supports every advanced course ahead: control theory, communications, digital signal processing, robotics, and aerospace sensing

Master the concepts, not just the calculations

How You're Evaluated

70%	18%	12%
Exams	Homework	Quizzes
Three midterms + final (25% each)	Problem-solving practice	Concept checks

Exam Bonus Structure

Ace all three midterms with A's?

Final exam waived

What We Value

Structured reasoning over final answers

Clear technical communication

Collaborative learning, individual work

Your Path to Success

01

Understand Structure

Don't memorize formulas, grasp why they work

02

Review Before Doing

Theory first, then tackle problems with confidence

03

Communicate Clearly

Treat solutions as technical documents: neat, logical, complete

04

Ask Early

Questions before exams > panic during exams

TA Information

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  - Senior in Electrical Engineering

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- Feel free to email me for questions regarding the course or grades.